

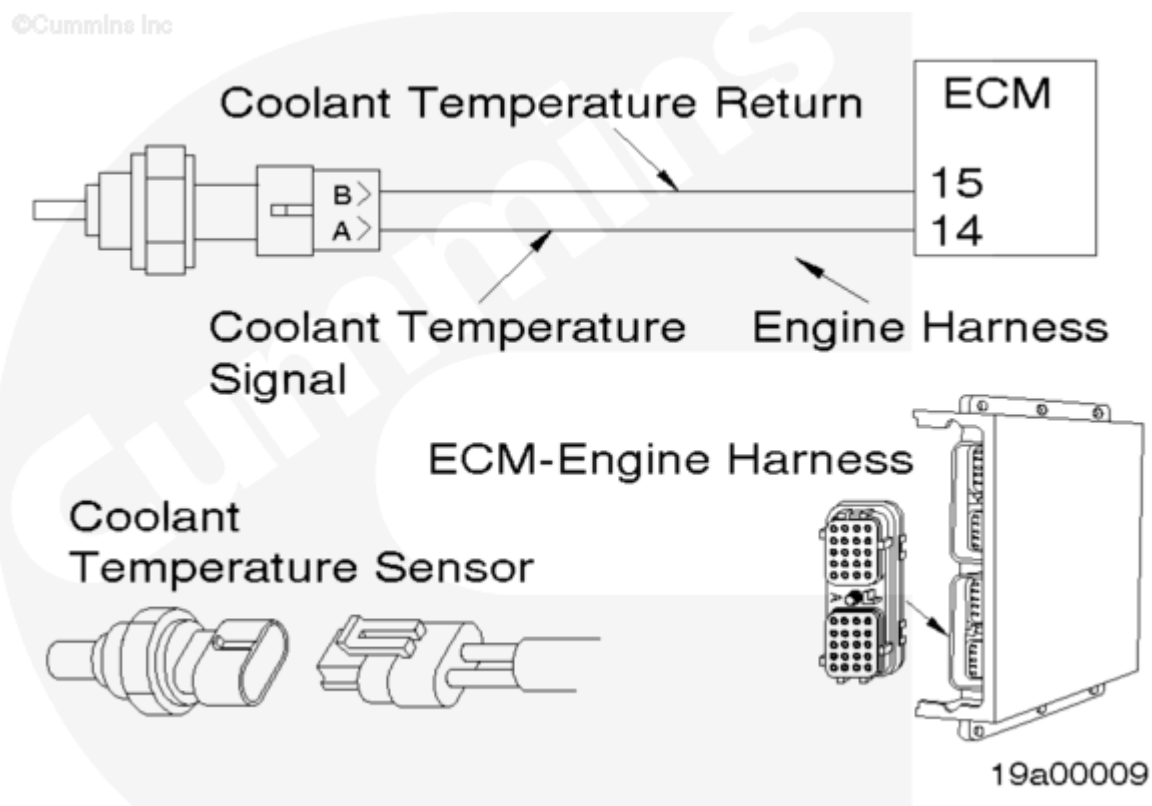
Fault Code: 146

Coolant Temperature - Engine Protection

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 146 PID(P): SPN: FMI: Lamp: SRT:	Engine coolant temperature has exceeded the warning threshold for high coolant temperature.	No effect on performance. Common Warning output is energized. Pre-High Coolant Temperature relay driver is energized.



Circuit Description

The CTS is used by the electronic control module (ECM) to monitor the temperature of the engine coolant. The ECM monitors the voltage on the signal pin and converts this to a temperature value. The coolant temperature is used by the ECM for the engine protection system and fueling

control.

Component Location

The CTS is located on the side of the thermostat housing.

Shoptalk

- Make sure the air flow through the radiator is **not** obstructed.
- The resistance of all the temperature sensors varies with the temperature. The reading that you observe should compare to the following table if the sensor is functioning properly.
- The threshold for the coolant temperature warning is adjustable with INSITE™, Part No. 3825145. Ensure the threshold is set to the appropriate value.

Temperature (° C)	Temperature [° F]	Resistance (ohms)
0	32	30k to 36k
25	77	9k to 11k
50	122	3k to 4k
75	167	1350 to 1500
100	212	600 to 675

Warnings and Cautions

WARNING

Wait until the coolant temperature is below 50° C [120° F] before removing the coolant system pressure cap or the CTS. Failure to do so can cause personal injury from heated coolant spray.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the sensor accuracy.	
	STEP 1A. Verify the sensor accuracy with a thermocouple or similar temperature probe.	Sensor reading is correct
STEP 2.	Clear the fault code.	

STEPS		SPECIFICATIONS
	STEP 2A. Disable the fault code.	Fault Code 146 inactive
	STEP 2B. Clear the inactive fault codes.	All faults cleared

STEP 1. Check the sensor accuracy.

STEP 1A. Verify the sensor accuracy with a thermocouple or similar temperature probe.

Conditions :

- Stop/Run switch in the "STOP" position.
- Controller in the diagnostic mode.

Action	Specification/Repair	Next Step
• Connect the temperature probe to the engine near the CTS. • Connect INSITE™, Part No. 3825145, to the equipment data link. • Compare the coolant temperature reading on the service tool monitor screen to the reading from the temperature probe. NOTE: If no temperature measuring device is available, then answer "OK" to this step.	Sensor reading is correct. Refer to the Base Engine Troubleshooting and Repair Manual for correct specifications.	2A
	Go to Fault code 145	Fault code 145

STEP 2. Clear the fault code.

STEP 2A. Disable the fault code.

Conditions : Connect all the components.		
Action	Specification/Repair	Next Step
- Connect all the components. - Start the engine and let it warm up to normal operating temperature to verify that the fault has been fixed.	Fault Code 146 inactive	2B
	Return to the troubleshooting steps or contact your local Cummins Authorized Repair Location if all the steps have been completed and checked again.	1A

STEP 2B. Clear the inactive fault codes.

Conditions : Connect all the components.		
Action	Specification/Repair	Next Step
Erase the inactive fault codes using INSITE™, Part No. 3825145. NOTE: The datalink connector is located on the right bank of the flywheel housing.	All faults cleared.	Repair complete
	Troubleshoot any remaining active fault codes.	Appropriate troubleshooting chart